

# CompositionMeasure for MaxDivergence

Michael Shoemate (@Shoeboxam)

This proof resides in “**contrib**” because it has not completed the vetting process.

Proves soundness of the implementation of **CompositionMeasure** for **MaxDivergence** in **mod.rs** at commit **f5bb719** (outdated<sup>1</sup>).

## 1 Hoare Triple

### Precondition

#### Compiler-verified

Types matching pseudocode.

#### Caller-verified

None

### Pseudocode

```
1 class CompositionMeasure(MaxDivergence):
2     def composability( #
3         self, adaptivity: Adaptivity
4     ) -> Composability:
5         if matches(adaptivity, Adaptivity.FullyAdaptive):
6             raise "fully-adaptive composition is not currently supported for max-divergence"
7         return Composability.Concurrent
8
9     def compose(self, d_mids: Vec[Self_Distance]) -> Self_Distance:
10         d_out = 0.0
11         for d_mid in d_mids:
12             d_out = d_out.inf_add(d_mid)
13         return d_out
```

### Postcondition

**Theorem 1.1.** `composability` returns `Ok(out)` if the composition of a vector of privacy parameters `d_mids` is bounded above by `self.compose(d_mids)` under `adaptivity` `adaptivity` and `out-composability`. Otherwise returns an error.

*Proof.* By the postcondition of **InfAdd** we have that  $\sum_i d\_mids_i \leq \text{compose}(d\_mids)$ .

| Adaptivity     | Sequential        | Concurrent        |
|----------------|-------------------|-------------------|
| Non-Adaptive   | Theorem 1[DMNS06] | Theorem 1.8[VW21] |
| Adaptive       | Theorem 1[DMNS06] | Theorem 1.8[VW21] |
| Fully-Adaptive | None              | None              |

<sup>1</sup>See new changes with `git diff f5bb719..141c5b6 rust/src/combinators/sequential_composition/mod.rs`

This table is reflected in the implementation of `composability` on line [2](#).

□

## References

- [DMNS06] Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. Calibrating noise to sensitivity in private data analysis. In Shai Halevi and Tal Rabin, editors, *Theory of Cryptography*, pages 265–284, Berlin, Heidelberg, 2006. Springer Berlin Heidelberg.
- [VW21] Salil Vadhan and Tianhao Wang. Concurrent composition of differential privacy, 2021.